Luenberger and Kalman Observers and Discrete Position Control of the Elbow Joint of a SCARA Robot

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***Abstract*—Two discrete state observers, a full-order predictor Luenberger observer and a linear Kalman filter, are designed and implemented to estimate the armature current and the speed of a Pololu 37D #4753 (50:1) gearmotor, measuring only the speed with an incremental encoder. The parametric model from the previous laboratory is tuned so that it draws $0.2\,\mathrm{A}$ and reaches the nominal speed at $12\,\mathrm{V}$, and it is refined with steps measured on the elbow joint of a SCARA robot. With the same model, a discrete PID position controller is designed directly in the z-plane by pole placement, with a reference prefilter and anti-windup. Everything runs on a Raspberry Pi Pico with a $10\,\mathrm{ms}$ sampling period and is operated from a web interface that allows changing the reference live. Both observers give consistent current estimates; under disturbances the Kalman filter tracks the speed with a $7.8\,\mathrm{rad/s}$ RMS error, against $34.8\,\mathrm{rad/s}$ for the Luenberger observer, and its covariance converges in three samples. Over 27 measured steps, the position loop settles in $0.74\,\mathrm{s}$ to $0.85\,\mathrm{s}$ with less than $1\,\%$ overshoot, against $0.65\,\mathrm{s}$ to $0.75\,\mathrm{s}$ in the design.**

***Index Terms*—Luenberger observer, Kalman filter, discrete control, pole placement, PID, DC motor, Raspberry Pi Pico.**

I. Introduction

In many applications not every variable of a system can be measured, either because no sensor is available or because measuring it is expensive. In a DC motor the speed is easily obtained with an encoder, but the armature current requires an additional sensor. A state observer reconstructs the unmeasured states from the system model, the input and the measured outputs [4].

This work implements the two observers proposed in the laboratory guide [1], [2]: the Luenberger observer, designed by pole placement with Ackermann's formula, and the linear Kalman filter, designed from the model and measurement uncertainty [5]. Unlike the guide, which uses a myRIO board and LabVIEW, the implementation runs on a Raspberry Pi Pico driving the elbow motor of a SCARA robot. Table I lists the four activities of the guide's procedure and where each one is reported. As additional work beyond the guide, a discrete-time position controller is designed with the same discrete model, so that the observers are also evaluated with the motor in closed loop (Sections VI and VIII-D).

Table I
Activities of the Laboratory Procedure

Activity of the guide [1]	Where
1. Tune the parametric model to the measured data	Sec. IV-B, Fig. 2, Table II
2. State space, simulation of current and speed, $0.2\,\mathrm{A}$ and nominal speed at $12\,\mathrm{V}$, comparison with the measurements	Sec. IV-B, Figs. 1 and 2
3. Luenberger observer from the speed measurement	Sec. V-A, VIII-A/B, Figs. 3–7
4. Linear Kalman filter from the speed measurement	Sec. V-B, VIII-A–C, Figs. 4–8
Additional: discrete position control	Sec. VI, VIII-D, Figs. 9, 10

II. Theoretical Framework

A. DC Motor Model

With the states $x = [i_a \quad \omega]^T$, the input $u = V_a$ and the measured output $y = \omega$, the motor equations are

$$\dot{i}_a = -\frac{R_a}{L_a} i_a - \frac{K_v}{L_a}\omega + \frac{1}{L_a} V_a, \tag{1}$$

$$\dot{\omega} = \frac{K_v}{J} i_a - \frac{\beta}{J}\omega, \tag{2}$$

that is, $\dot{x} = A_c x + B_c u$, $y = Cx$ with

$$A_c = \begin{bmatrix} -\frac{R_a}{L_a} & -\frac{K_v}{L_a} \\ \frac{K_v}{J} & -\frac{\beta}{J} \end{bmatrix}, \quad B_c = \begin{bmatrix} \frac{1}{L_a} \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}. \tag{3}$$

B. Discretization

Since the Pico holds the voltage constant between samples, the actuator is a zero-order hold with sampling period T_s:

$$A = e^{A_c T_s}, \qquad B = \int_0^{T_s} e^{A_c \tau}\, d\tau\, B_c, \tag{4}$$

$$x_k = A x_{k-1} + B u_{k-1}, \qquad y_k = C x_k. \tag{5}$$

Each continuous pole s maps to $z = e^{sT_s}$; a mode much faster than T_s ends up at $z \approx 0$.

C. Luenberger Observer

The full-order predictor observer uses the input and the measurement from the previous sample:

$$\hat{x}_k = (A - LC)\hat{x}_{k-1} + L\,y_{k-1} + B\,u_{k-1}. \tag{6}$$

The estimation error evolves as $e_k = (A - LC)e_{k-1}$, so L is chosen to place the eigenvalues of $A - LC$ inside the unit circle. With Ackermann's formula,

$$L = \Phi(A)\,\mathcal{O}^{-1}\begin{bmatrix}0\\1\end{bmatrix}, \qquad \mathcal{O} = \begin{bmatrix}C\\CA\end{bmatrix}, \tag{7}$$

where $\Phi(z)$ is the desired characteristic polynomial.

D. Linear Kalman Filter

Starting from $\hat{x}_0$ and P_0, each sample performs a prediction and a correction [5]:

$$\hat{x}_k^- = A\hat{x}_{k-1}^+ + Bu_{k-1}, \qquad P_k^- = AP_{k-1}^+A^T + Q, \tag{8}$$

$$K_k = P_k^-C^T(CP_k^-C^T + R)^{-1}, \tag{9}$$

$$\hat{x}_k^+ = \hat{x}_k^- + K_k(y_k - C\hat{x}_k^-), \qquad P_k^+ = (I - K_kC)P_k^-, \tag{10}$$

where Q represents the model uncertainty and R the measurement variance. P_k is the covariance of the estimation error. To follow its evolution with a single number,

$$p_k = \sqrt{P_{11,k}^2 + P_{22,k}^2}, \tag{11}$$

the norm of its diagonal, is used. If p_k grows without bound the filter diverges; if it decreases and settles, the filter converges. With constant A, Q and R, P_k follows the Riccati equation and converges to a fixed value that does not depend on the measurements [5]; this is why it does not go to zero.

E. Discrete Control by Pole Placement

Let the position plant be $G(z) = N(z)/D(z)$ and the controller

$$C(z) = \frac{U(z)}{E(z)} = \frac{q_0z^2 + q_1z + q_2}{(z-1)(z+g)}, \tag{12}$$

which is a discrete PID with a filtered derivative: the pole at $z = 1$ gives the integral action and the pole at $z = -g$ filters the derivative. The closed loop has the characteristic polynomial

$$(z-1)(z+g)D(z) + N(z)\,(q_0z^2 + q_1z + q_2) = \Phi_c(z), \tag{13}$$

which is linear in g, q_0, q_1, q_2. If $D(z)$ is second order, matching coefficients with the fourth-order desired polynomial $\Phi_c(z)$ gives four equations in four unknowns [6]. The desired poles come from continuous-time specifications: the dominant pair $s = -\sigma \pm j\omega_d$, with $\sigma = 4/t_s$ and $\omega_d = \sigma\sqrt{1-\zeta^2}/\zeta$, maps to the z-plane as $z = e^{sT_s}$. Solving (12) for u gives the difference equation

$$u_k = (1-g)u_{k-1} + g\,u_{k-2} + q_0e_k + q_1e_{k-1} + q_2e_{k-2}. \tag{14}$$

If the already saturated voltage is stored in u_{k-1} and u_{k-2}, the integral part does not accumulate while the actuator is at its limit (anti-windup).

III. Experimental Setup

The elbow motor of a SCARA RRP robot is used: a 12 V Pololu 37D #4753 gearmotor with a 50:1 gearbox [3], which drives the forearm through a 3:1 pulley. The Hall encoder gives 64 counts per revolution of the motor shaft. The speed is computed as the position difference between two samples divided by T_s, so its resolution is $2\pi/(64\,T_s) \approx 9.8\,\mathrm{rad/s}$. At the elbow, the angular resolution is $2\pi/(64{\cdot}150) \approx 0.65\,\mathrm{mrad}$. The motor is driven by a BTS7960 H-bridge with a 20 kHz PWM from a Raspberry Pi Pico, which reads the encoder with a PIO state machine.

The firmware runs a loop every $T_s = 10\,\mathrm{ms}$ that measures, updates both observers, computes the voltage (fixed in the open-loop experiments, or given by the position controller) and sends the data over USB. Since the elbow has a limited travel (about $3.7\,\mathrm{rad}$), the firmware homes against the limit switch, cuts the voltage before either end of travel and stops the motor if it stalls. The prismatic axis is kept braked. When the elbow accelerated, its reaction torque turned the shoulder; therefore the shoulder is held by its own PD loop ($u = 30\,e - 1.0\,\omega$, 4 V limit), which runs in a $10\,\mathrm{ms}$ interrupt independent of the elbow loop and releases if the shoulder moves more than $0.15\,\mathrm{rad}$ away.

On the PC, a Python server keeps the serial port open and a web page sends the commands (*home*, voltage steps or `pos REF T`) and plots the angle, the speed, the estimated currents, the voltage and p_k live. During a position run, the reference can be changed without stopping the loop (`r REF`). Each run is saved to a CSV file.

IV. Motor Model

A. Laboratory 1 Model

The previous parametric identification gave the parameters in the Lab 1 column of Table II, with an armature resistance of $72.7\,\Omega$. With that model the motor reaches $0.084\,\mathrm{A}$ and $558\,\mathrm{rad/s}$ ($107\,\mathrm{rpm}$ at the output) at 12 V, far from the $0.2\,\mathrm{A}$ and $200\,\mathrm{rpm}$ specified by the manufacturer. K_v and β were consistent with the datasheet, but R_a was about 30 times larger than the value derived from the stall current ($5.5\,\mathrm{A}$ at 12 V).

B. Model Tuning

R_a was set to $12/5.5 = 2.18\,\Omega$, and the two steady-state conditions of the guide were imposed, $i_a = 0.2\,\mathrm{A}$ and $\omega = \omega_{nom} = 1047\,\mathrm{rad/s}$ ($10\,000\,\mathrm{rpm}$ at the motor shaft), which gives

$$K_v = \frac{12 - R_a\,i_a}{\omega_{nom}}, \qquad \beta = \frac{K_v\,i_a}{\omega_{nom}}. \tag{15}$$

J and L_a were then fitted by least squares to reproduce three steps of 4, 6 and 8 V measured on the elbow. The resulting J includes the forearm inertia reflected to the motor shaft. Table II summarizes the parameters.

With these values,

$$A_c = \begin{bmatrix}-3924.1 & -19.86\\ 3935.3 & -0.752\end{bmatrix}, \quad B_c = \begin{bmatrix}1798.6\\0\end{bmatrix}, \tag{16}$$

Table II
MOTOR PARAMETERS

Parameter	Lab 1	Tuned
R_a (Ω)	72.7	2.18
L_a (H)	6.62×10^{-2}	5.56×10^{-4}
K_v (V s/rad)	1.06×10^{-2}	1.104×10^{-2}
J (kg m^2)	8.55×10^{-7}	2.81×10^{-6}
β (N m s)	1.59×10^{-6}	2.11×10^{-6}
i_a at 12 V (A)	0.084	0.200
ω at 12 V (rad/s)	558	1047
τ_{mech} (ms)	273	48

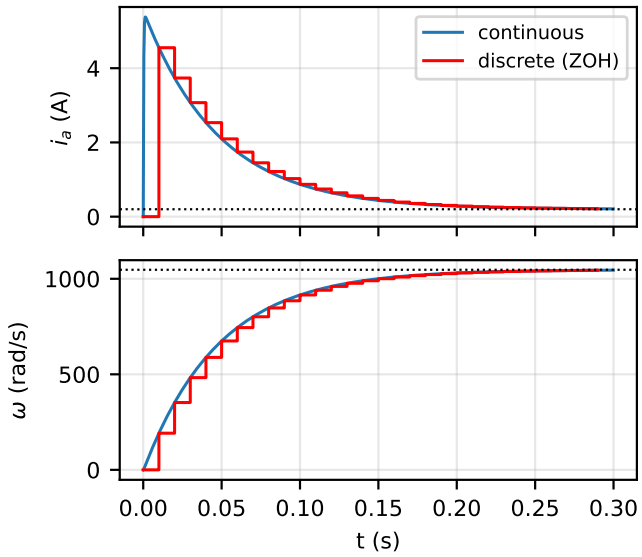


Figure 1. Response of the tuned model to 12 V: the current settles at 0.2 A and the speed at 1047 rad/s.

and with $T_s = 10\,\text{ms}$,

$$A = \begin{bmatrix} -0.00419 & -0.00415 \\ 0.8233 & 0.8166 \end{bmatrix}, \quad B = \begin{bmatrix} 0.3793 \\ 15.991 \end{bmatrix}. \tag{17}$$

The discrete plant poles are $z \approx 0$ (electrical mode, $\tau_e = L_a/R_a \approx 0.25\,\text{ms} \ll T_s$) and $z = 0.812$ (mechanical mode). Fig. 1 verifies the nominal condition and Fig. 2 compares the model with the data.

The tuned model reproduces the measured time constant, but overestimates the speed by about 40 rad/s at 4 V. The measured steady-state speeds (305, 496 and 687 rad/s) grow by about 96 rad/s per volt, and extrapolated to 12 V they give $\approx 1070\,\text{rad/s}$, close to the nominal speed. However, the line does not pass through the origin: there is a dead zone of about 0.8 V caused by the dry friction of the gearbox and the arm, which a linear model cannot represent.

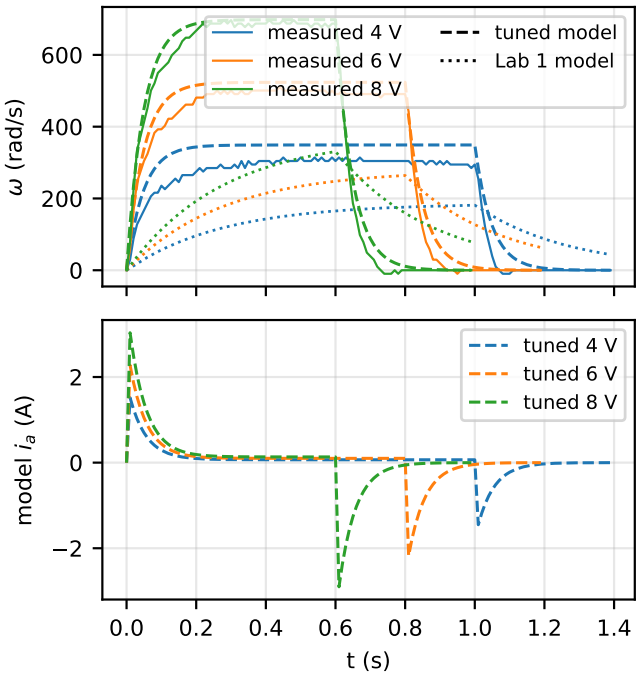


Figure 2. Steps of 4, 6 and 8 V measured on the elbow, compared with the Lab 1 model (dotted) and with the tuned model (dashed).

V. OBSERVER DESIGN

A. Luenberger

The observability matrix

$$\mathcal{O} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0.8233 & 0.8166 \end{bmatrix}, \qquad \det \mathcal{O} = -0.8233, \tag{18}$$

has rank 2, so the system is observable from the speed measurement alone. The initial design, with poles at $z = 0.3$ and 0.4, produced negative current estimates down to -3.5 A with the real data (Fig. 3). The reason is that the electrical pole of the plant is already at $z \approx 0$: moving it to 0.3 makes the current mode of the observer slower than that of the plant, and the small model error (the friction) accumulates in the current. The poles $z = \{0,\ 0.2\}$ were therefore chosen, which keep the electrical pole and speed up the mechanical one, giving

$$\Phi(z) = z^2 - 0.2z, \qquad L = \begin{bmatrix} -0.00312 \\ 0.6124 \end{bmatrix}. \tag{19}$$

B. Kalman

The measurement variance was taken from the steady-state data of the steps, between 26 and 45 $(\text{rad/s})^2$, and $R = 35$ was used. For the model, an uncertainty of 10 mA in the current and 10 rad/s in the speed was assumed, and the initial estimate was given a larger uncertainty in the current, which is not measured:

$$Q = \begin{bmatrix} 10^{-4} & 0 \\ 0 & 100 \end{bmatrix}, \quad P_0 = \begin{bmatrix} 0.1 & 0 \\ 0 & 100 \end{bmatrix}. \tag{20}$$

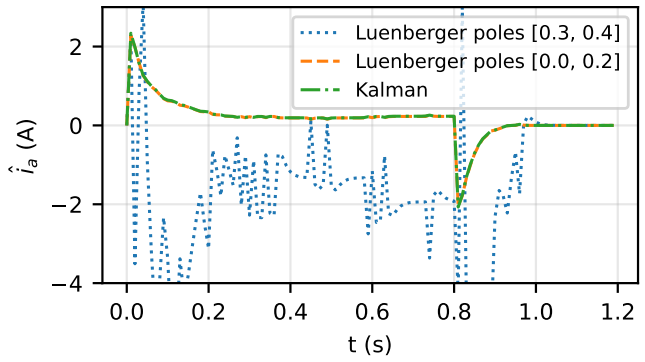


Figure 3. Estimated current with the data of the 6 V step: with the poles $\{0.3, 0.4\}$ the Luenberger observer gives negative currents; with $\{0, 0.2\}$ it matches the Kalman filter.

The gain converges to $K = [-5.98 \times 10^{-4} \quad 0.771]^T$. No retuning was needed with these values.

VI. Position Controller Design (Additional Work)

A. Discrete Plant

The elbow angle, $\dot{\theta} = \omega/150$ (50:1 gearbox and 3:1 pulley), is added to the model of Section IV, and the result is discretized with a zero-order hold. The transfer function from voltage to angle is third order, but the electrical pole ($z \approx 10^{-17}$) cancels with a zero at $z = -0.0012$, which leaves

$$G(z) = \frac{\theta(z)}{U(z)} = \frac{5.372 \times 10^{-4}\,(z + 1.029)}{(z - 1)(z - 0.8124)}. \qquad (21)$$

The pole at $z = 1$ is the integrator (the position is the integral of the speed) and the one at $z = 0.8124$ is the mechanical mode. The zero at $z = -1.029$ is a sampling zero introduced by the hold; since it lies outside the unit circle, the design does not cancel it.

B. Controller

The specifications $\zeta = 0.7$ (about 5 % overshoot) and $t_s = 0.5\,\text{s}$ give $\sigma = 8\,\text{s}^{-1}$, $\omega_d = 8.16\,\text{rad/s}$ and the dominant pair $z = 0.9231\,e^{\pm j0.0816} = 0.920 \pm 0.075j$. The two remaining poles were placed at $z = 0.6$, faster than the dominant pair. Solving (13) with (21) gives

$$q_0 = 139.71, \qquad q_1 = -247.97, \qquad (22)$$

$$q_2 = 110.03, \qquad g = -0.3027, \qquad (23)$$

and it was verified that the closed-loop poles lie exactly at the requested values.

C. Reference Prefilter

Since q_0 is large, a 1 rad reference step would ask for more than 100 V. With the saturation at $\pm 6\,\text{V}$, in simulation the voltage jumped from $+6$ to $-6\,\text{V}$ between two samples, with current peaks of $\pm 3\,\text{A}$, because the saturated loop is no longer linear. The reference therefore goes first through a second-order prefilter outside the loop,

$$F(z) = \left(\frac{(1-\alpha)z}{z-\alpha}\right)^2, \qquad \alpha = 0.93, \qquad (24)$$

with time constant $-T_s/\ln\alpha \approx 0.14\,\text{s}$. The prefilter does not change the closed-loop poles: it only smooths the reference that enters the loop. In simulation, with $V_{max} = 6\,\text{V}$, steps from 0.5 to 2.2 rad settle in 0.65 s to 0.75 s without overshoot.

VII. Implementation

The observer matrices and the controller coefficients are computed in Python and written to header files that the firmware compiles. Since $C = [0 \;\; 1]$, the innovation is a scalar and the Kalman filter needs no matrix inversion. The Luenberger observer step is:

```
// x(k) = (A - L C) x(k-1) + L y(k-1) + B u(k-1)
float e = y_ant - x[1];
float x0 = AD[0]*x[0] + AD[1]*x[1] + BD[0]*u_ant + LG[0]*e;
float x1 = AD[2]*x[0] + AD[3]*x[1] + BD[1]*u_ant + LG[1]*e;
```

the Kalman filter correction:

```
float s = Pm[3] + RK; // C P- C' + R
float K0 = Pm[1]/s, K1 = Pm[3]/s; // gain
float e = y - x1; // innovation
x[0] = x0 + K0*e; x[1] = x1 + K1*e;
```

and the position controller, with the prefilter and the difference equation (14):

```
rf1 = a*rf1 + (1-a)*ref; // prefilter F(z)
rf = a*rf + (1-a)*rf1;
float e = rf - art; // error in rad
float u = (1-g)*u1 + g*u2 + q0*e + q1*e1 + q2*e2;
u = limitar(u, -vmax, vmax); // u1, u2 keep the saturated u
e2 = e1; e1 = e;
```

The 10 ms loop includes the measurement, both observers, the controller, the protections and the data transmission, including p_k from (11).

VIII. Results

A. Observers: Simulation with Measured Data

Before the implementation, both observers were run on the measured steps with the same algorithm as the firmware (Fig. 4). Over the three steps the speed RMS error was 12–14 rad/s for Luenberger and 2.6–3.1 rad/s for Kalman. The estimated steady-state currents agreed between both methods (0.27, 0.23 and 0.19 A at 4, 6 and 8 V).

B. Observers on the Pico in Open Loop

Fig. 5 shows a 5 V step estimated in real time. Both observers follow the measured speed, and the estimated current shows the start-up peak (1.9 A), settles at ≈0.25 A and becomes negative during braking.

Under voltage changes of 4, 8, 3, −6 and −3 V (Fig. 6), the observers follow the speed in every segment, including the direction reversal. The estimated current shows a peak at each change and reaches −3 A when the direction reverses, which is the expected behavior of a DC motor.

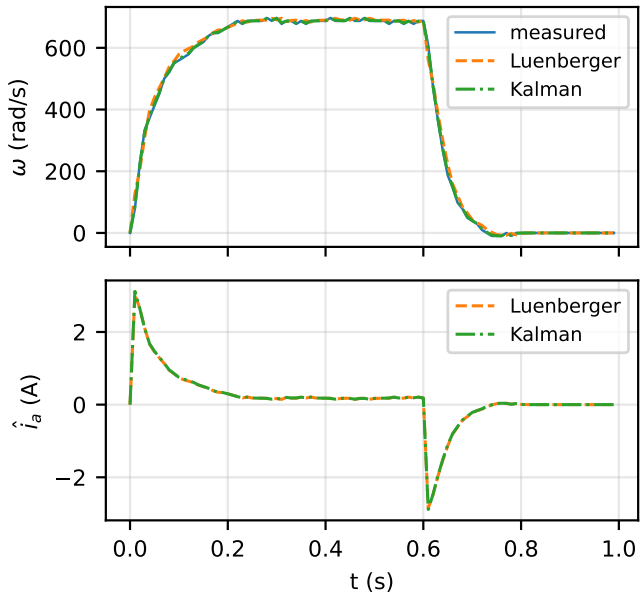


Figure 4. Simulation of the observers on the measured 8 V step.

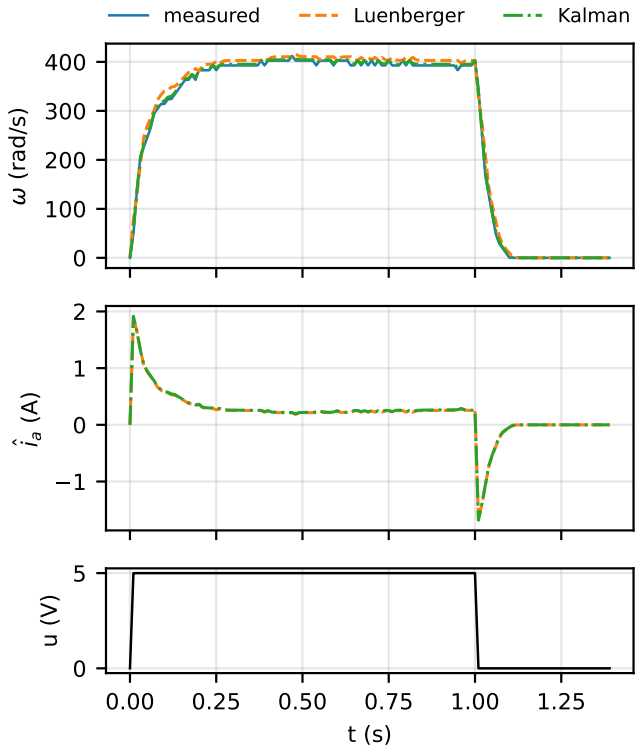


Figure 5. Implementation on the Pico: 5 V step for 1 s.

For the disturbance test, the elbow moves back and forth at ±3.5 V for 6 s, reversing at 0.5 and 1.4 rad, while the forearm is braked by hand (Fig. 7). Each time the load is applied, the measured speed drops and the estimated current rises from ≈0.4 A to 1.0–1.3 A. That value approaches the stall current

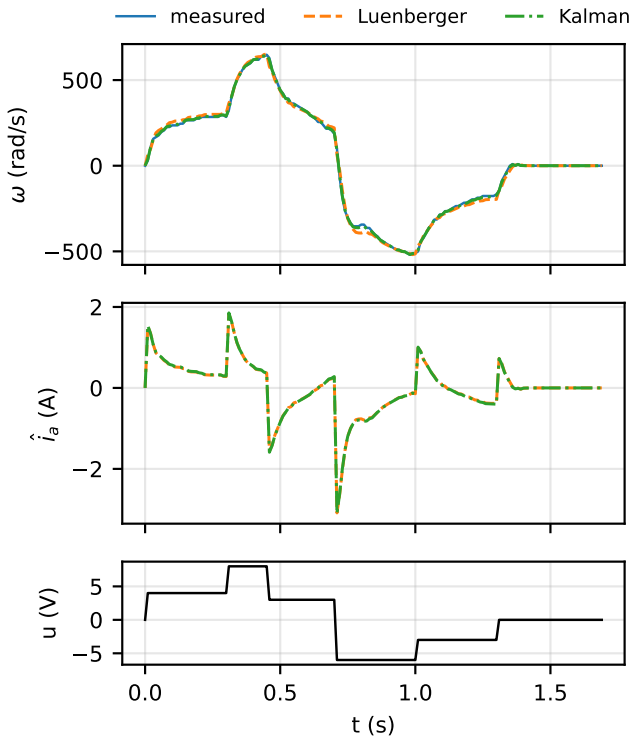


Figure 6. Voltage changes: 4, 8, 3, −6, −3 and 0 V.

Table III
RMS Error of the Estimated Speed (rad/s)

Test	Luenberger	Kalman
4, 6 and 8 V steps (simulation)	12–14	2.6–3.1
Load disturbance (Pico)	34.8	7.8

at 3.5 V ($3.5/R_a \approx 1.6$ A). Table III summarizes the errors.

C. Convergence of the Kalman Covariance

Fig. 8 shows p_k computed on the Pico. It starts at $p_{-1} = 100$ (the initial estimate P_0), drops to 28.93 at the first correction and to 27.06 at the second, and from the third sample on it stays at $p = 26.99$, which corresponds to $P_{11} = 5.1 \times 10^{-4}$ A^2 and $P_{22} = 26.99\,(\text{rad/s})^2$. The value does not change with the reference steps or with the load, as expected from a linear filter with constant Q and R: the filter quickly converges to a steady uncertainty, which is not zero because the model and the measurement always have some error. Since $P_{22} \gg P_{11}$, p_k reflects almost only the speed uncertainty.

D. Position Control of the Elbow (Additional Work)

Fig. 9 compares a step from 0.5 to 2.0 rad measured on the elbow with the simulation of the design. The shape of the response matches: in both cases the voltage saturates at 6 V for about 0.3 s and the elbow arrives without overshoot. The difference is in steady state: on the robot the controller holds ≈2 V to keep the position, and in simulation 0 V.

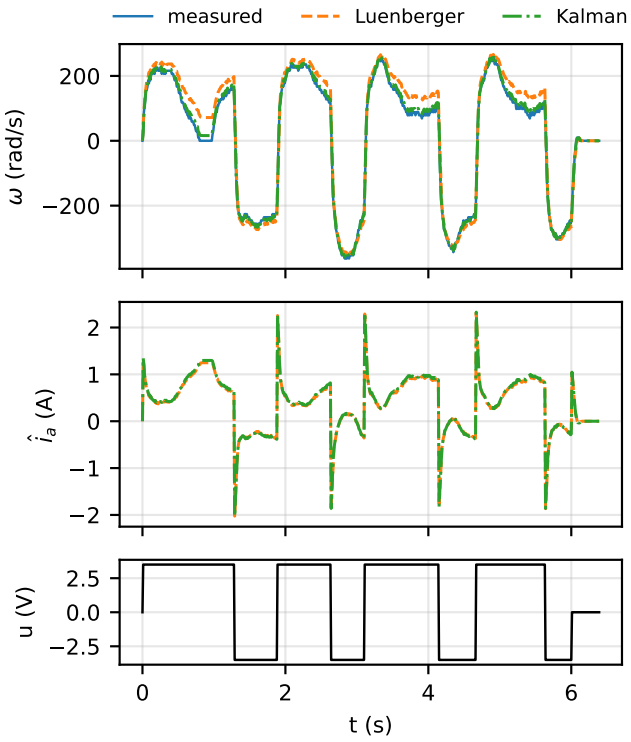


Figure 7. Disturbance: the elbow moves back and forth at $\pm 3.5\,\mathrm{V}$ and is braked by hand three times.

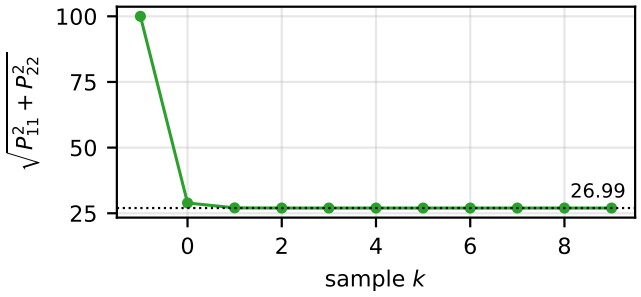


Figure 8. Norm of the diagonal of the Kalman filter covariance on the Pico; $k = -1$ corresponds to P_0.

Fig. 10 shows a run in which the reference was changed live from the interface, between 0.5, 1.0 and $2.0\,\mathrm{rad}$. The observers keep working in closed loop: the estimated current shows a $1.5\,\mathrm{A}$ to $1.8\,\mathrm{A}$ peak when starting and a negative one when braking at each change, and between changes it stays at $0.4\,\mathrm{A}$ to $0.7\,\mathrm{A}$ with a sign that depends on the position, like the voltage held by the controller.

Table IV summarizes 27 complete steps from 0.5 to $1.73\,\mathrm{rad}$ measured in five runs with the shoulder held. The settling time (2 % band) was $0.74\,\mathrm{s}$ to $0.85\,\mathrm{s}$, about $0.1\,\mathrm{s}$ longer than in simulation, and the overshoot never exceeded $9.1\,\mathrm{mrad}$ (less than 1 %).

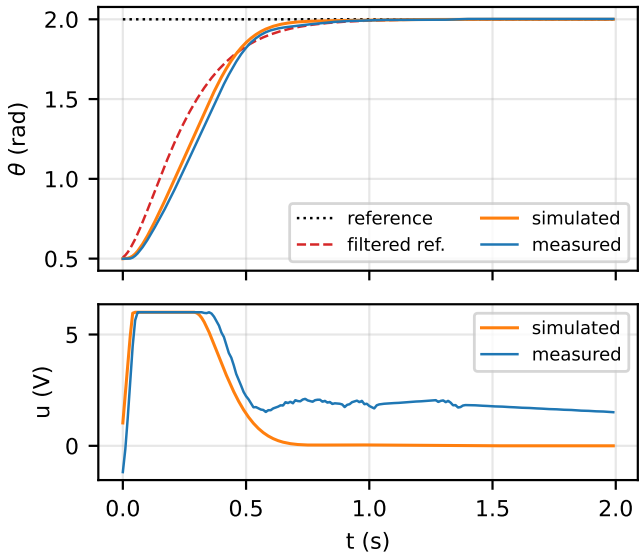


Figure 9. Step from 0.5 to $2.0\,\mathrm{rad}$ with $V_{max} = 6\,\mathrm{V}$: simulated design and measurement on the elbow.

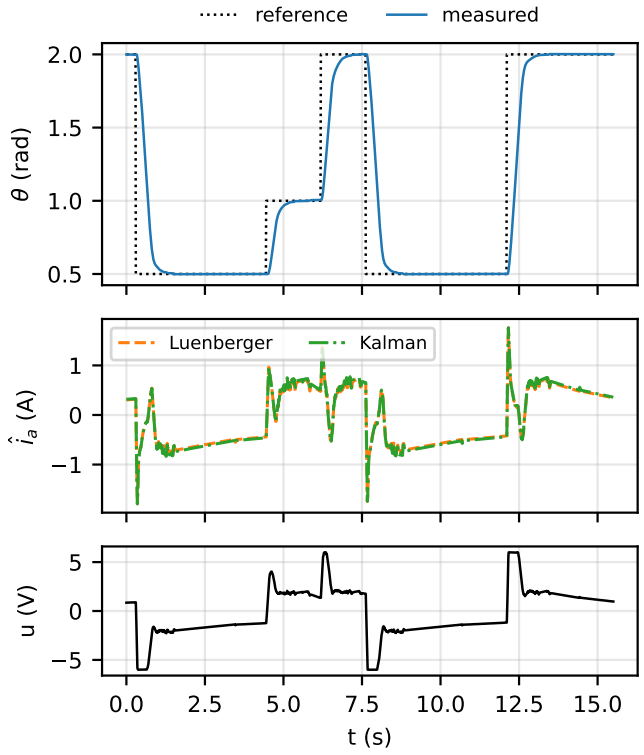


Figure 10. Position control of the elbow with live reference changes: angle, current estimated by both observers and applied voltage.

IX. Discussion

Tuning the model was necessary: the Lab 1 model did not meet the nominal condition because its armature resistance was about 30 times larger than expected. By fixing K_v and β

Table IV
POSITION CONTROL PERFORMANCE ($V_{max} = 6\,\mathrm{V}$)

	Design (simulation)	Measured on the elbow	
		Not saturated	Saturated
Steps	–	18	9
Amplitude (rad)	0.5–2.2	0.50–0.72	1.00–1.73
t_s 2 % (s)	0.65–0.75	0.75–0.85	0.74–0.77
Max. overshoot (mrad)	0	4.7	9.1
Mean final error (mrad)	0	2.1	1.2
Max. final error (mrad)	0	4.7	3.9

from the steady-state conditions and fitting only J and L_a, the model meets the manufacturer's specification and reproduces the measured dynamics. The remaining error is due to dry friction, which is not linear.

For the Luenberger observer, the choice of poles turned out to be decisive. With $T_s = 10\,\mathrm{ms}$ the electrical mode of the plant is practically instantaneous, and an observer pole far from $z = 0$ makes the current estimate very sensitive to model errors. The Kalman filter, on the other hand, computes its gain from Q and R without choosing poles by hand, and it behaved well from the first design. Its covariance converges in three samples and remains constant; for it to react to changes in the operating conditions, Q or R would have to vary in time, as in an adaptive filter.

With the motor braked, the estimated speed does not reach zero (Fig. 7): neither model includes the load torque, so the observer expects the motor to turn whenever voltage is applied. Kalman suffers less from this bias because its gain on the speed (0.77) gives more weight to the measurement. The same effect appears in closed loop: to hold the elbow still the controller applies $\approx$2 V, with opposite sign at 0.5 and at $2.0\,\mathrm{rad}$, which indicates an external torque toward the middle of the travel, probably from the cables. The integral action compensates that torque without error, but the observers interpret it as current with the motor at rest. Adding the load torque as a third state would solve both issues.

The z-plane design held on the robot: the shape of the response and the overshoot match the simulation, and the settling time is only $0.1\,\mathrm{s}$ longer. That difference and the final error of a few milliradians are due to dry friction, which the linear model does not include: near the reference the voltage is small and the elbow stops before arriving. In one run, the integral action kept raising the voltage against that friction with a $4.5\,\mathrm{mrad}$ error until it exceeded $4\,\mathrm{V}$, and the stall protection stopped the run even though the elbow had arrived. The prefilter was essential: without it, the high q_0 makes the voltage swing between both saturation limits at every reference change.

A limitation of the setup also appeared in practice: when the direction was reversed with the motor at high speed, the current peak and the energy returned by the motor shut down the power supply. The open-loop experiments were therefore limited to reversals of $\pm 6\,\mathrm{V}$ or less in the middle of the elbow travel, and the position controller was saturated at 6 V.

X. CONCLUSIONS

- With a single speed measurement, both observers reconstruct the armature current consistently, in steady state, in transients, under load and with the motor in closed loop.
- Under disturbances, the Kalman filter tracks the speed about four times better than the Luenberger observer (7.8 versus $34.8\,\mathrm{rad/s}$ RMS) and filters the encoder quantization noise. Its covariance converges in three samples to a constant value.
- The Luenberger design depends on the chosen poles. Modes much faster than the sampling period should be kept at $z \approx 0$; the same criterion allowed reducing the position plant to second order.
- The discrete PID designed by pole placement in z met its design on the robot: over 27 steps it settled in $0.74\,\mathrm{s}$ to $0.85\,\mathrm{s}$ with less than $1\,\%$ overshoot and a mean final error of $1.8\,\mathrm{mrad}$. The prefilter and the anti-windup were necessary because of the actuator saturation.
- The Raspberry Pi Pico runs both observers, the controller and the data acquisition in a $10\,\mathrm{ms}$ loop without difficulty, which makes it a low-cost alternative to the myRIO board.

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